
EARLY WARNING SYSTEM FOR OFFLINE STUDENT DROPOUTS BASED ON EDUCATIONAL DATA MINING TECHNIQUES

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ABSTRACT

In the field of education, student dropout is still a major problem, especially in traditional classroom settings. In order to forecast and reduce the danger of student dropouts, this study suggests creating an Early Warning System (EWS) that makes use of Educational Data Mining (EDM) approaches. Early in the academic cycle, the system detects at-risk pupils by examining academic, demographic, behavioural, and attendance data. To create prediction models, the study uses machine learning techniques like Random Forest, Logistic Regression, and Decision Trees. The findings show that incorporating EDM into teaching methods greatly enhances the capacity to recognise and assist students who are at danger.

Keywords: Machine Learning, Offline Classroom, Educational Data Mining, Early Warning System, and Dropout Prediction.

I. INTRODUCTION

Dropout from educational institutions negatively impacts individual futures and educational ecosystems. In offline classrooms, where immediate digital footprints are minimal, identifying dropout risks is challenging. Traditional methods rely on manual observation, which is often subjective and delayed. Educational Data Mining (EDM) offers a proactive solution by analyzing historical and real-time student data to develop Early Warning Systems (EWS) that can predict dropouts before they occur.

II. LITERATURE REVIEW

Although comparatively few studies concentrate on offline classrooms, prior research has shown the value of EDM in online learning contexts. Academic risks can be accurately predicted using student information such as attendance, grades, involvement, and socioeconomic status. Students have been successfully categorised according to their risk levels through the use of machine learning algorithms such as Decision Trees and Logistic Regression.

III. OBJECTIVES

- To determine the main causes of student dropouts in traditional classroom settings. To create
- Prediction models using data mining techniques in order to identify at-risk students early.
- To use model outputs to create and assess an early warning system.

IV. METHODOLOGY

4.1 Data Collection: Over the course of two academic years, data was gathered from offline classroom settings in a semi-urban educational institution. The dataset consists of:

Academic results

Records of attendance

Reports on behaviour

Background in terms of socioeconomic status

Frequency of parent-teacher interactions

4.2 Data Preprocessing: Standard methods were used to encode the dataset and remove any missing values. Numerical values on the scale were normalised.

4.3 Feature Selection: To determine the most pertinent predictors of dropout risk, correlation analysis and recursive feature elimination (RFE) were employed.

4.4 Model Construction: Three models underwent training: Classifier Using Decision Trees Regression using Logistic The Random Forest Classifier Metrics like accuracy, precision, recall, and F1-score were used to assess these models.

4.5 Development of Early Warning Systems (EWS): To visualise high-risk kids based on model predictions, a rule-based dashboard was created. Academic mentors were alerted to take action.

V. FINDING

The Random Forest model fared better than the Decision Tree (82%) and Logistic Regression (79%), with an accuracy of 87%. Attendance and academic success were the most significant factors. The EWS identified 85% of dropouts at least two months before they actually stopped participating.

VI. DISCUSSION

According to the study, dropout risk in offline classes can be accurately predicted using EDM approaches. Teachers and administrators can use the EWS as a tool to help them make decisions. Based on these projections, early intervention can enhance educational outcomes and student retention.

VII. CONCLUSION

One possible strategy for addressing student dropouts in offline situations is the implementation of an Early Warning System employing EDM approaches. Teachers can improve student success by moving from reactive to proactive intervention tactics by utilising both historical and present data.

VIII. UPCOMING PROJECTS

Future studies could use AI-powered tools to combine real-time classroom observations with psychological aspects. The models' generalisability will be put to the test by expanding to different educational contexts (such as rural and multilingual ones).

IX. REFERENCES

- [1] Bowers, A. J. (2021). Early warning systems and indicators of dropping out of upper secondary school: the emerging role of digital technologies. *OECD Digital Education Outlook*, 173.
- [2] Chung, J. Y., & Lee, S. (2019). Dropout early warning systems for high school students using machine learning. *Children and Youth Services Review*, 96, 346-353.
- [3] Isiaka, R. M., Babatunde, R. S., Ajao, F. J., & Abdulsalam, S. O. (2019). A machine learning approach to dropout early warning system modeling. *International Journal of Advanced Studies in Computers, Science and Engineering*, 8(2), 1-12.
- [4] Mi, C. (2019). Student performance early warning based on data mining. *International Journal of Performability Engineering*, 15(3), 822.
- [5] Pecuchova, J., & Drlik, M. (2024). Enhancing the Early Student Dropout Prediction Model Through Clustering Analysis of Students' Digital Traces. *IEEE Access*.
- [6] Deeva, G., De Smedt, J., & De Weerd, J. (2022). Educational sequence mining for dropout prediction in MOOCs: Model building, evaluation, and benchmarking. *IEEE Transactions on Learning Technologies*, 15(6), 720-735.
- [7] Duong, H. T. H., Tran, L. T. M., To, H. Q., & Van Nguyen, K. (2023). Academic performance warning system based on data driven for higher education. *Neural Computing and Applications*, 35(8), 5819-5837.
- [8] Sáiz-Manzanares, M. C., Rodríguez-Díez, J. J., Díez-Pastor, J. F., Rodríguez-Arribas, S., Marticorena-Sánchez, R., & Ji, Y. P. (2021). Monitoring of student learning in learning management systems: An application of educational data mining techniques. *Applied Sciences*, 11(6), 2677.
- [9] Rahul, & Katarya, R. (2023). A Systematic Review on Predicting the Performance of Students in Higher Education in Offline Mode Using Machine Learning Techniques. *Wireless Personal Communications*, 133(3), 1643-1674.
- [10] Laili, U. F., Tanzeh, A., Efendi, N., & Gufron, M. (2024). K-Means Clustering Method On Academic Advising Management And Early Detection Of Student Dropout (Sequential Explanatory Mixed Method Study At UIN Sunan Ampel Surabaya And IAIN Kediri). *Int. J. Educ. Res. Soc. Sci*, 5(2), 335-346.